NSO8062 Geofüüsikaline andmetöötlus

Loeng 4: Andmete silumine

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Ajakava ja teemad

Nädal	Teema	Õppejõud
1	Sissejuhatus ainesse	S-T. Luik
2	Erinevad andmete tüübid ja lihtsamad tegevused nendega.	S-T. Luik
3	NetCDF failide loomine, muutmine jne.	S-T. Luik
4	Keri andmed. Profiilide visualiseerimine, andmete eemaldamine, silumine, gradientide leidmine.	S-T. Luik
5	Andmete 1D interpoleerimine ja korrelatsioonid, statistikud jne.	S-T. Luik
6	Tuuleandmed ja nende visualiseerimine	S-T. Luik
7	2D andmed, interpoleerimine, visualiseerimine (nt kaart)	
8	Kuumalained	
9	TBD	
10	TBD	
11	TBD	
12	TBD	
13	TBD	
14	TBD	
15	Konsultatsioon	
16	**Eksam**	
17		
18		
19	10/06/2026 kevadsemestri eksamite sooritamise lõpptähtaeg	
20	15/06/2026 kevadsemestri lõpp	

Sissejuhatus

- Kasutame andmeid, mis on kogutud Keri saare lähedalt profileerijaga.
  - Keri avamere püsijaam on merre paigaldatud mõõtesüsteem, mis on ühendatud veealuse kaabli abil Keri saare tugipunktiga ja on võimeline registreerima erinevate okeanograafiliste parameetrite vertikaalseid profiile mere pinnast kuni põhjani ning mõõdetud andmeid reaalajas edastama.
- *Julian Day* – päeva number aastas, 1 … 365/366
- *Halokliin* ehk soolsuse hüppekiht – piirkond veesambas, kus sooluse väärtused muutuvad järsult.
- Andmete silumine – “käsitsi” ja erinevate sisseehitatud funktsioonide abil.
- Alguses teooria, pärast harjutused.

Smoothing data - The smoothdata function

smoothed_data = smoothdata(data, ’approach’, window size, NaN-handling)

SD = smoothdata(D, ‘movmean’, 5, ‘omitnan’)

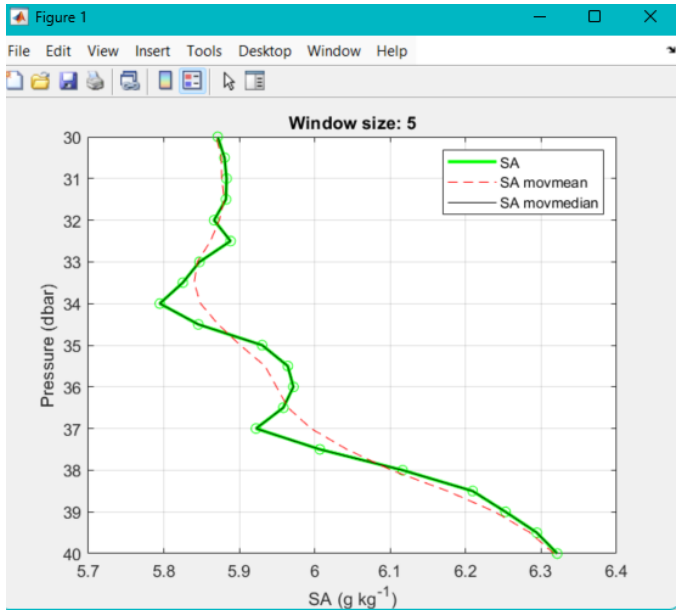
Window size:

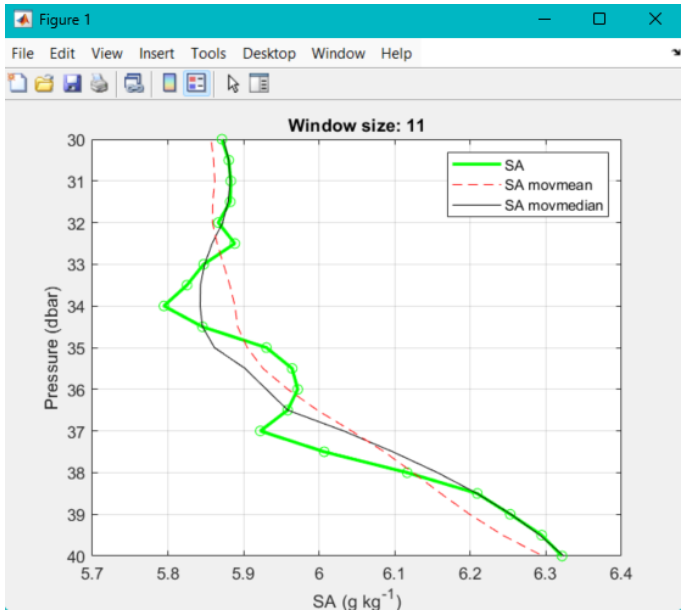
Point Location	Depth	Included in the Average?
$i-3$	3.5m	No
$i-2$	**4.0m**	**Yes (Start)**
$i-1$	**4.5m**	**Yes**
i	**5.0m**	**Center Point**
$i+1$	**5.5m**	**Yes**
$i+2$	**6.0m**	**Yes (End)**
$i+3$	6.5m	No

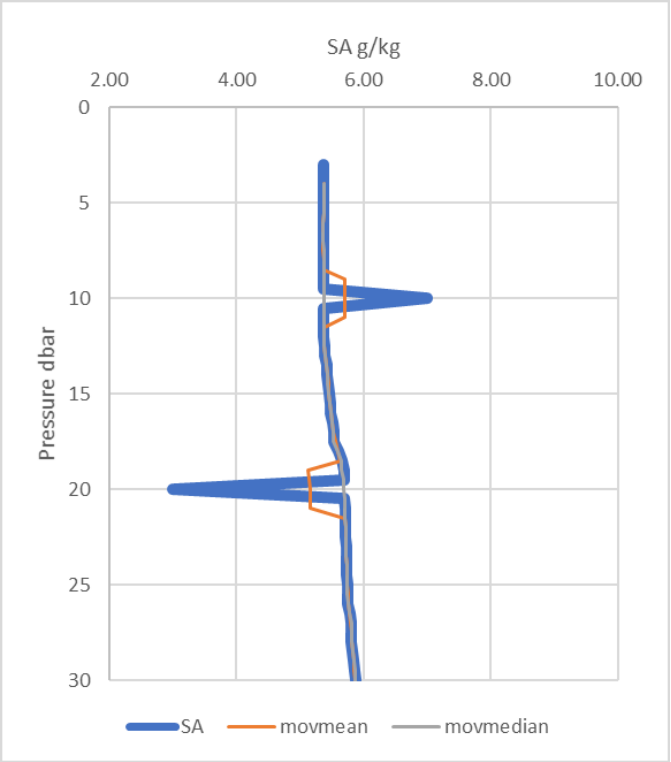
Smoothing data - The smoothdata function

movmean - Average over each window. Sensitive to outliers (one bad spike can pull the mean a lot. Use when measurements are clean and you mainly want to suppress regular oscillations.

movmedian - Median over each window. A few extreme bad points won't strongly affect the result. Use when you expect spikes/outliers.



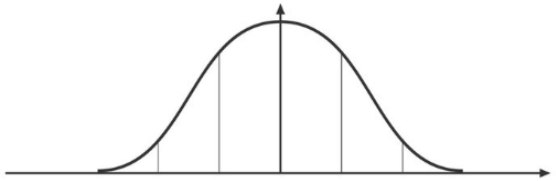




Smoothing data – The smoothdata function

gaussian - Gaussian-weighted average over each window.

While a Moving Average treats every data point in the window equally (giving them all a "weight" of 1), a Gaussian-weighted average gives more importance to the point at the center of the window and less to the points at the edges.

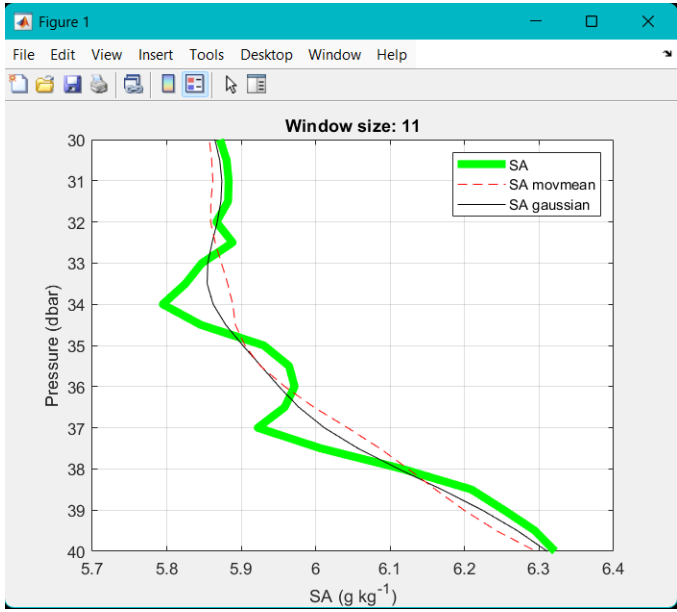


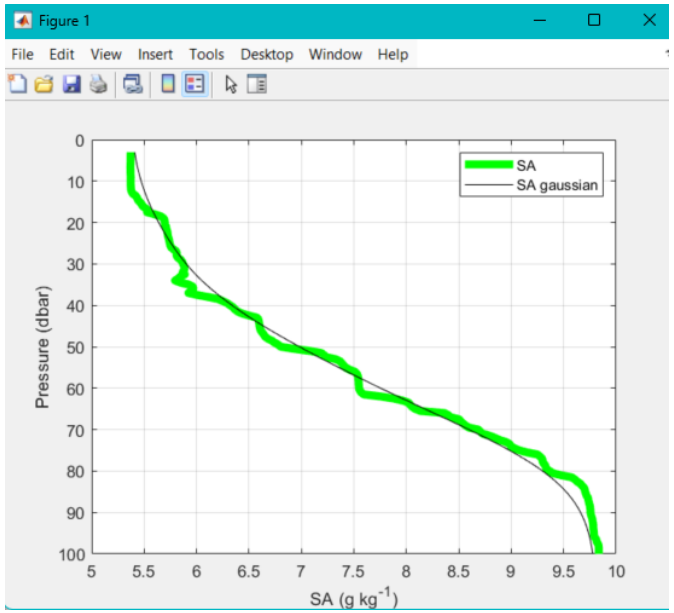
The Logic: How it Weights Data

Imagine a bell curve (Gaussian distribution) centered on your target depth.

- Data exactly at that depth is multiplied by the peak of the curve.
- Data 1 meter away is multiplied by a smaller value.
- Data at the very edge of the window is multiplied by almost zero.

The "width" of this bell curve is defined by the Standard Deviation (σ). By default, MATLAB sets the window size to roughly 6 σ.





Smoothing data – The smoothdata function

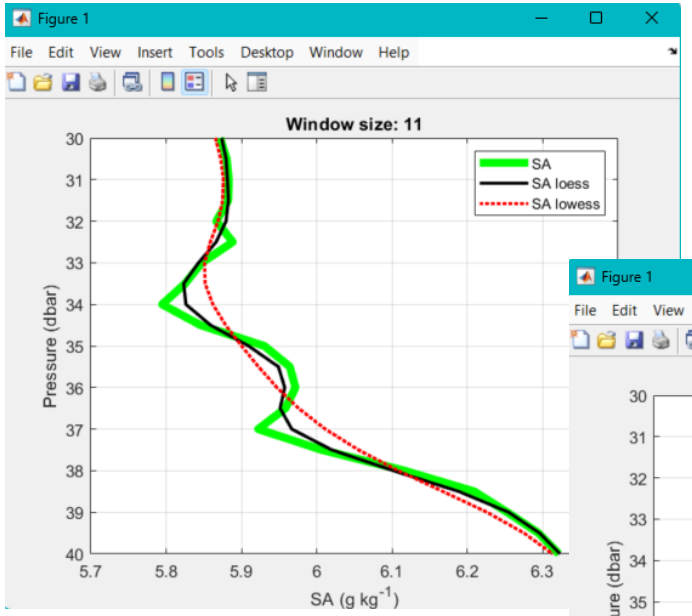
1. lowess - Linear regression over each window. This method can be computationally expensive.

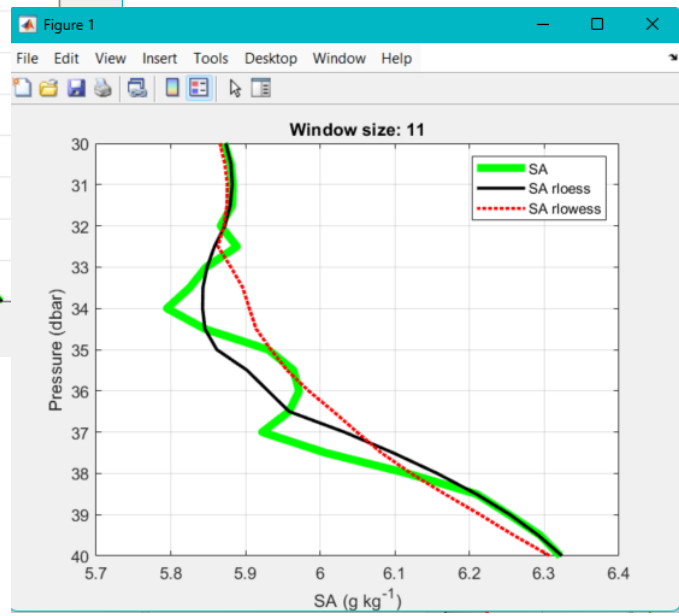
2. loess - Quadratic regression over each window. This method is slightly more computationally expensive than "lowess".

3. rlowess - Robust linear regression over each window. This method is a more computationally expensive version of the method "lowess", but it is more robust to outliers.

4. rloess - Robust quadratic regression over each window. This method is a more computationally expensive version of the method "loess", but it is more robust to outliers.

Feature	Linear Regression	Quadratic Regression
Line Shape	Straight	Curved (Parabola)
Rate of Change	Constant	Variable (Accelerating/Decelerating)
Number of Bends	Zero	One
Extrapolation	Usually safer for short distances	Dangerous (curves can shoot to infinity quickly)





Smoothing data - The smoothdata function

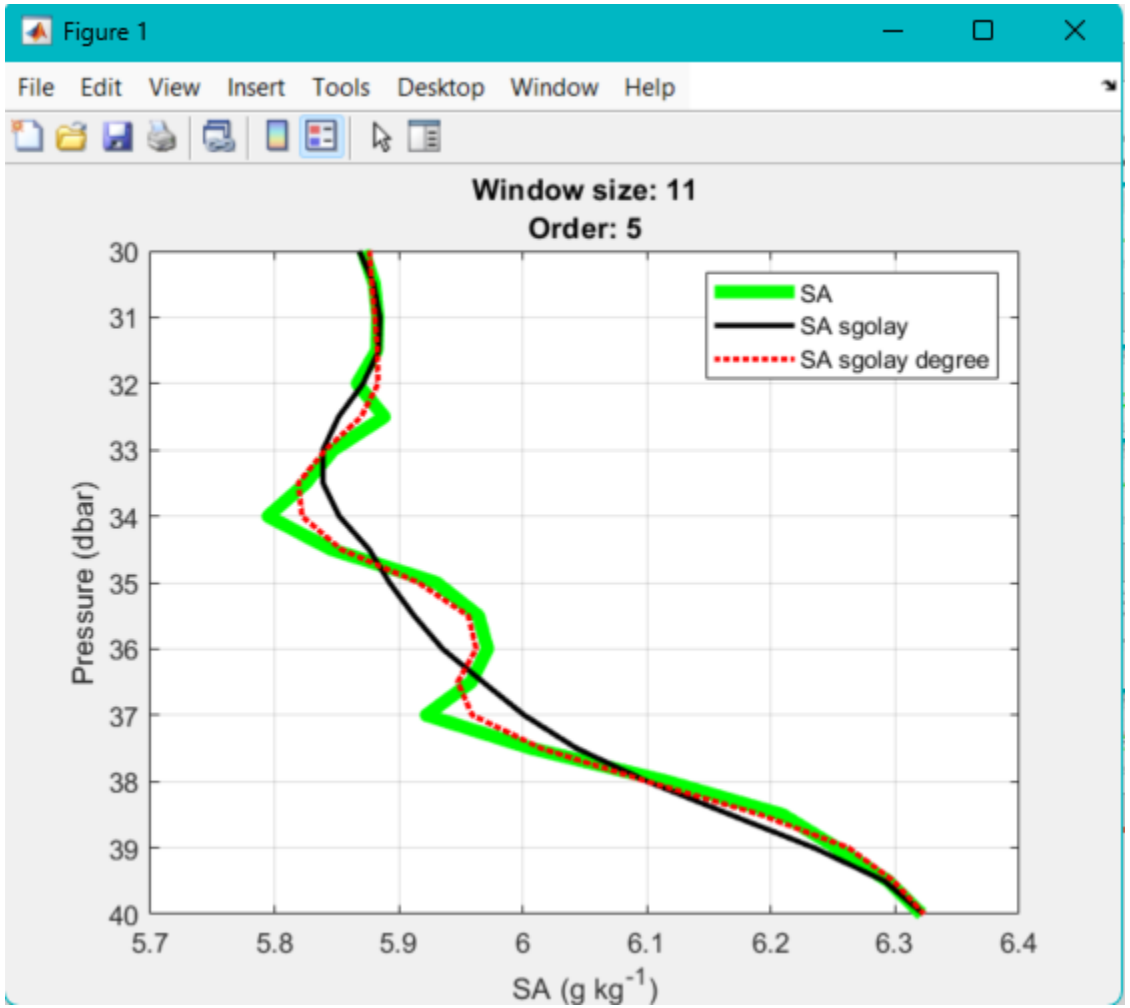
sgolay - Savitzky-Golay filter, which smooths according to a quadratic polynomial that is fitted over each window of A.

This method can be more effective than other methods when the data varies rapidly.

Default degree for the polynomial is 2.

In oceanographic profiles, the polynomial order determines how "flexible" the smoothing line is:

- Order 2 (Quadratic): Good for general noise reduction where the data is relatively smooth.
- Order 3 (Cubic): Better at "following" the curve of a sharp thermocline or halocline. It captures the change in curvature (the "knees" of the profile) more accurately than a 2nd-order fit.
- Higher Orders (4+): Risk re-introducing noise by over-fitting to the "wiggles" you are trying to remove.



Smoothing data - The smoothdata function

If your data has "glitches" (random points that are way off the scale), movmean, loess, and lowess will all be pulled toward those glitches, creating a "bump" in your smooth line. Use the Robust versions (rlowess or rloess) or movmedian to tell MATLAB to essentially ignore those outliers.

Lowess vs. Loess: The difference is the degree of the polynomial:

- Lowess (Linear): Better for data that moves in straight-ish segments.
- Loess (Quadratic): Better for data with lots of bends and waves. It "hugs" the curves more tightly.

The Savitzky-Golay (sgolay) Exception

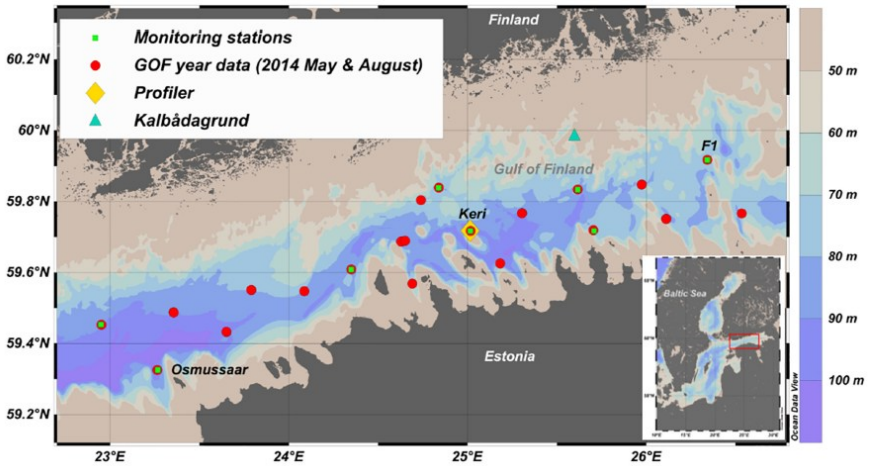
- Unlike the others, sgolay is designed specifically to preserve the higher-frequency moments of a signal. While a movmean will "flatten" a sharp peak, sgolay attempts to keep the peak at its original height while smoothing the "fuzz" on the sides.

Method	Type	Weighting	Best Use Case	Key Characteristic
movmean	Moving Average	Uniform	General noise reduction.	Simple, but blurs sharp features and is sensitive to outliers.
movmedian	Moving Median	Uniform	Removing outliers (spikes).	Extremely robust; ignores salt-and-pepper noise entirely.
gaussian	Weighted Moving Average	Gaussian	Smoothing high-frequency jitter.	Gives more weight to the center; smoother results than movmean.
lowess	Linear Local Regression	Weighted (Tri-cube)	General trend estimation.	Fits a 1st-degree polynomial (line) to each window.
loess	Quadratic Local Regression	Weighted (Tri-cube)	General trend estimation.	Fits a 2nd-degree polynomial; handles curvature better than lowess.
rlowess	Robust lowess	Iterative Weights	Data with extreme outliers.	A "hardened" version of lowess that ignores points with high residuals.
rloess	Robust loess	Iterative Weights	Curvy data with outliers.	The "hardened" version of loess; very computationally expensive but clean.
sgolay	Savitzky-Golay Filter	Non-weighted	Preserving peaks/shapes.	Excellent for signal processing (e.g., spectroscopy) where peak height matters.

Lecture 4. Exercise 1.

For this exercise, let's use profiles from Keri station.

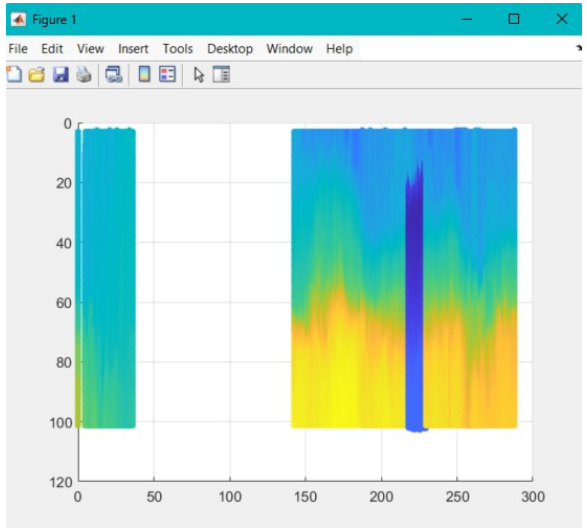
- Keri_2020.mat



Browse data, create a figure of temporal change of salinity profiles.

Julian Days – juda_UTC
Depth – depth_m
Salinity – SA_gkg

scatter, Ydir reverse, grid on



Remove bad data:

Using Keri data, identify profiles, where salinity at max depth is < 7 g kg^{-1}.
Remove these profiles.
Remove data from the beginning of the year also.
Plot data again. Add names to axes and a title. Add colorbar. Save figure as .png.
Save table as .mat. (Keri_2020_clean.mat)

Add descriptions to your code and save code.

Lecture 4. Exercise 2.

Keri_2020_clean.mat

- Plot all salinity profiles (x = SA, y = pressure, color by Julian Day).
- Add colorbar and labels based on Julian Day. Add labels to axes.

Select a subset of data, where Julian Day > 270

- Smooth data using indexing (mean, over 2.5 m) and find halocline. Use variables SA_gkg, pressure_dbar.
  - Halocline is a layer, where vertical salinity gradient > 0.07 g kg^{-1} m^{-1} (according to Liblik and Lips, 2011). Find halocline only where pressure is > 40 dbar.
- Visualize results (SA values in halocline layer).

Get deepest point of halocline & oxygen (do_mgl) at bottom for each Julian Day.

- Put these variables and unique Julian Days into a table and save said table – results_lec4_ex2.mat

Add descriptions to your code and save code.

Lecture 4. Exercise 2. (continued)

1. Select a subset of data, where Julian Day > 270. Use variables SA_gkg, pressure_dbar.

2. Smooth data using indexing (mean, over 2.5 m).

   Create a ‘manual mean’ (don’t use smoothdata function). Write a code that gets a mean value for depth *i* by averaging values from *i-2* to *i+2*.

   SA_smooth(i) = mean(SA_gkg(i-2):SA_gkg(i+2))

Point Location	Depth	Included in the Average?
$i-3$	3.5m	No
$i-2$	**4.0m**	**Yes (Start)**
$i-1$	**4.5m**	**Yes**
i	**5.0m**	**Center Point**
$i+1$	**5.5m**	**Yes**
$i+2$	**6.0m**	**Yes (End)**
$i+3$	6.5m	No

3. Find halocline.

   Halocline is a layer, where vertical salinity gradient > 0.07 g kg^{-1} m^{-1} (according to Liblik and Lips, 2011). Find halocline only where pressure is > 40 dbar.

   NEW: (formula for data with a 0.5 dbar vertical step)

   3.1. SA_gradient(i) = (SA_smooth(i+2)-SA_smooth(i-2)) / (pressure_dbar(i+2)-pressure_dbar(i-2))

   3.2. halocline = SA_gradient > 0.07 & pressure_dbar > 40

Lecture 4. Exercise 3.

Keri_2020_clean.mat

Select one profile and smooth it with 2 different options.

- Use smoothdata function with different approaches, window sizes and visualize data.

Add descriptions to your code and save code.

Lecture 4. Exercise 4.

Keri_2020_clean.mat

FIG 1:

Smooth all profiles with 2 different options. Plot results.

Find halocline from smoothed profiles and add to plots as a line.

FIG 2 & 3:

Visualize the difference of measured profiles and smoothed profiles

1. x = Julian Day, y = Pressure, z = Salinity - Salinity_smooth1
2. x = Julian Day, y = Pressure, z = Salinity - Salinity_smooth2

Add descriptions to your code and save code.